Rutu Mahesh Ghatge

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SUMMARY

AI engineer with 2+ years of experience building and shipping LLM-powered products for real users. Built the frontend and backend of Vybie, a skincare coaching app, led the team implementation of its doctor-consultation Goals feature, and led backend development of the live Formulynx.ai platform. Experience with RAG pipelines, LangChain/LangGraph, MCP integrations, real-time chat and payments, delivered with stakeholders and under tight deadlines. Earlier computer vision work at ARDE DRDO. Comfortable presenting to clients and open to travel.

SKILLS

LLMs & Agents: LangChain, LangGraph, RAG, MCP, Langfuse (LLM evaluation and monitoring), Prompt Engineering, LLM Integration (LLaMA, DeepSeek), Dialogflow CX, Vertex AI

Voice: ElevenLabs, Deepgram, faster-whisper, Edge TTS

Backend & Data: Python, FastAPI, Flask, Django, WebSockets, Socket.IO, MongoDB, MySQL, SQLite3, Docker, AWS, C++

Frontend: React, Tailwind CSS, admin panels and analytics dashboards

Integrations: Razorpay, Shopflo, Meta Analytics

ML & Computer Vision: TensorFlow, PyTorch, Keras, OpenCV, Scikit-learn, YOLO, DeepSORT, CNN, LSTM, Kalman Filters, NLP

Tools: Git & GitHub, Cursor, Antigravity, Windsurf, Claude Code, Power BI, Agile & Scrum

Communication: Client presentations, requirements gathering, cross-functional collaboration, public speaking

EXPERIENCE

AI Engineer **June 2025 – Present**

Techs & Tomes (Skin Beyond Borders Pvt Ltd) *Pune*

- **End-to-end ownership:** Took AI features at SkinBB from non-technical business requirements through to shipped products, then evaluated and monitored LLM performance and cost using Langfuse.
- **Vybie (skincare coaching app):** Built the complete frontend and backend on my own and designed the credit system.
- **Skin Friendliness Index:** Combines live temperature, UV index, humidity and AQI with the user's profile data to generate personalised skin alerts.
- **Time Machine:** Compares two selfies and reports improvement or deterioration as a percentage. Six compare kits: Glam Check, Then vs Now, Quikee (for products claiming instant results), Dusk→Dawn, Dawn→Dusk and a 28-Day Trial.
- **Label Looker:** v1.0 analyses ingredients from a scan. v2.0 is embedded in the product pages of the SkinBB app and website and checks product fit against the user's profile along with ingredient analysis.
- **Chatbot voice:** Added speech-to-text and text-to-speech using Deepgram and ElevenLabs.
- **Goals (doctor consultations):** Led planning and implementation, as a team effort, of a feature where doctors connect with users through subscription plans: doctor panel, Razorpay payments, and live chat on WebSockets/Socket.IO monitored from the admin panel.
- **SkinBB admin panel:** Built deep analytics for B2B and B2C, brand settlements and invoice management. Integrated YouTube and Instagram content to make the website dynamic through the Get Social feature.
- Worked in a team on the frontend of almost all SkinBB website pages, the admin panel and the brand admin panel (React, Tailwind CSS), including API integration under tight deadlines.
- **Integrations:** Worked with MCP, Razorpay, Shopflo and Meta Analytics across SkinBB projects.
- **Formulynx.ai:** Led backend development of the live platform: APIs, database architecture and AI integration for real-time formulation analysis. Cleaned and enhanced 40,000+ raw ingredients data, creating a high quality dataset.
- Deployed a multilingual skincare chatbot using LangChain and RAG, reducing average query resolution time.
- Designed Skin Vision, a privacy-preserving AI facial skin analyser with budget-aware recommendations, integrated via React and Tailwind.
- Built an AI video translator and summariser (faster-whisper, Edge TTS; 50+ videos) and web scrapers that collected 40,000+ structured product records.

AI Intern **Nov 2024 – May 2025**

ARDE, DRDO *Pune*

- Designed a custom CNN for aerial object detection using GELAN activation and KMeans-based anchor box optimisation for small moving targets.

- Developed a hybrid Kalman Filter + LSTM model for real-time drone tracking with 99%+ predictive accuracy.
- Built a PDF-based RAG chatbot using LLaMA and DeepSeek to answer technical queries across 500+ defence documents.
- Researched drone-following-drone navigation using YOLO, DeepSORT, SLAM and D* path planning.
- Received a letter of recommendation (LOR) from ARDE, DRDO on completion of the internship.

Smart Traffic Management Intern **Feb 2025 – Mar 2025**
Infosys Springboard *Remote*

- Developed ANPR and ATCC systems (YOLO, OCR, Flask, React) with helmet detection, heatmaps, and crosswalk and signal violation detection; delivered all milestones in a 6-week Agile sprint cycle.

Development Intern **June 2024 – Nov 2024**
Standard Learner *Remote*

- Worked on the platform development of a mock test platform with active student users, in direct contact with the founder.
- Implemented new features, resolved production issues, and collaborated on chatbot development and AI model integration.

Other internships and training (2024), remote: Data Science Intern, CodeClause; Web Development Intern, TechnoHacks Edutech; Business Analyst Training, Finlatics.

PROJECTS

- **Investment Intelligence:** Multi-agent financial platform with a custom orchestrator routing to Portfolio, Analysis, News and Jargon agents. FastAPI, MongoDB, Gemini Pro, React.
- **RahiAI – Tourism Chatbot:** Full-stack chatbot for personalised recommendations and itinerary planning using real-time data APIs. React, FastAPI, LangGraph, MongoDB.
- **SafeSurf AI – Malicious Website Detector:** Browser extension for real-time protection using a Scikit-learn classifier with the VirusTotal API (94%+ accuracy). Django, Selenium, Playwright.
- **Multiformat Deepfake Detection:** Multimodal classifier for image, voice and text (91%+ accuracy). React, Flask.
- **AI integration consultation – Greenaray Labs (biotech):** With a faculty member during college, held discovery discussions with the lab's non-technical team on where AI could fit into their workflows.
- **Human Emulated Robotic Hand:** IoT flex-sensor glove driving a robotic replica; published in IJARSCT and presented at ICMETET.

RESEARCH

- **Published:** Human Emulated Robotic Hand (IJARSCT; presented at ICMETET), research on affordable neuro-prosthetics.
- **Published:** Leveraging LLMs for Explainable and Personalized Facial Skin Analysis Scopus-indexed publication: Research paper on utilizing fine-tuned LLMs and vision models to analyze skin features, producing dermatological guidelines under resource boundaries. Presented at ICETT-2026
- **Ongoing:** Theory of Mind in Multi-Agent RL; early PCOD/PCOS prediction using multimodal AI; multi-criteria decision making (Fuzzy-DEMATEL, VIKOR, TOPSIS) with Mechanical Engineering faculty.

CERTIFICATIONS & ACHIEVEMENTS

Certifications: Google AI Essentials, Introduction to Generative AI, Badge & Webhook (Google Cloud Skills Boost), SQL and Relational Databases (IBM), Foundations of Cybersecurity (Google), 45+ Infosys Springboard certificates.

Achievements: 1st place, State-level Hackathon (Ajeenkya DY Patil School of Technology); 1st place, DATAthon (ISBM COE); 1st place, Project Competition (ISB&M Bangalore); 1st place, Idea Hackathon (ISBM Kolkata); NPTEL Elite + Silver Medal (top 5%).

Activities: Coordinator, IoT Tech-lab and Research Centre; Member, Technical Placement Cell and Coding Club; Mentee, Infosys Springboard Pragati Program.

EDUCATION

B.E. Artificial Intelligence and Data Science **2022 – 2026**
ISBM College of Engineering, Savitribai Phule Pune University, Pune *SGPA: 8.62 / 10*

HSC 2022, Loyola Junior College, Pune: 73.33% | SSC 2020, Modern High School, NCL Campus, Pune: 93.40%

Languages: English, Hindi, Marathi